FLYGUARD AI

NEXT-GENERATION AML INTELLIGENCE

Explainable Graph & Behavioral Intelligence for Money-Mule Detection

Stopping multi-hop illicit financial flows without drowning AML teams in false positives.

ARCHITECTURE STACK FastAPI • XGBoost + Isotonic • NetworkX • TreeSHAP • Cytoscape.js

MARKET BREAKDOWN

The Mule Detection Crisis

90%–95% False Positives

Static rule alerts overwhelm investigators.

<60 Minute Flight

Instant rails scatter illicit deposits rapidly.

$2T Global Laundering

Regulators reject unexplainable black-box AI.

Core Impact: Legitimate SMBs get frozen while layered mule rings go undetected.

UPI · FedNow · RTP · SEPA

ARCHITECTURAL FLAWS

Why Existing Defenses Collapse

Legacy Static Rules

Blind to Topology: Misses circular rings and splitters.

Threshold Evasion: Mules operate just below dollar caps.

High Operational Cost: Freezes high-volume merchant SMBs.

Standard Black-Box ML

Target Leakage: Account ID overfitting inflates benchmarks.

Temporal Lookahead: Training leaks future transaction dates.

No SAR Filing: Regulators reject unexplainable predictions.

- Required: Topology-aware models with human-verifiable evidence chains.

PLATFORM OVERVIEW

FlyGuard Investigative Command Center

26 Anti-Leak Features

Velocity, burst, and graph centrality.

Calibrated Risk Index

0–100 monotonic risk index.

TreeSHAP Explainability

Plain-English evidence summaries.

Multi-Hop Graph Tracer

Interactive 1–3 hop ring tracing.

Full Case Dossiers & Tamper-Evident Audit Trails Included

12-PAGE COMMAND CENTER

RIGOROUS BENCHMARK

3-Tier Behavioral Simulation

200,000 Transactions across 20,000 Synthetic Accounts

seed=42 · Zero Target Leakage



Legitimate (79%)

Payroll, retail savings, and P2P transfers.

Hard Negatives (15%)

High-volume SMBs and payment collectors.

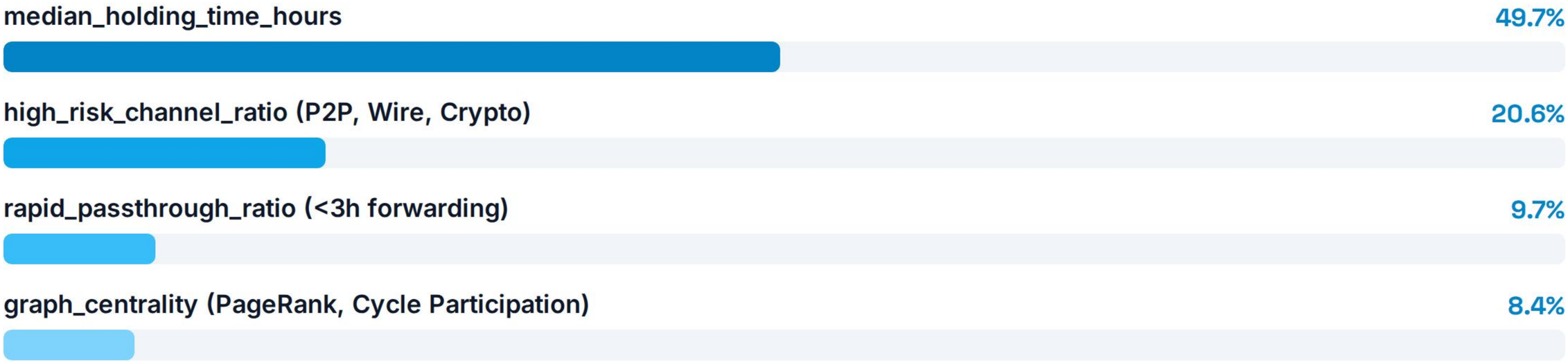
Money Mules (6%)

Funnels, circular rings, and burst accounts.

Anti-Leakage Protocol: GroupShuffleSplit on scenario IDs with strict temporal cutoff enforcement.

FEATURE IMPORTANCE

Predictive Behavioral Signals



Isotonic Calibration maps raw probabilities into calibrated triage bands: Low (<35) · Mod (35–65) · High (65–85) · Critical (>=85)

EXPLAINABILITY LAYER

TreeSHAP & Counterfactual Lab

TreeSHAP Attribution

Transparent log-odds drivers per account.

- **Holding Time <0.5h (+2.4 log-odds)**
- **Instant P2P Rail Ratio >90% (+1.8 log-odds)**

"What-If" Counterfactuals

Simulate parameter adjustments interactively.

Baseline Score		Simulated (+36h hold)
94.2	→	**18.5**

Provides defensible evidence narratives directly ingestible into SAR filings.

GRAPH INTELLIGENCE

Multi-Hop Network Graph Tracer

NetworkX + Cytoscape.js

Sub-second browser rendering of 1–3 hop rings.

Circular Cycle Detection

Surfaces recursive rings up to 5 hops deep.

Shortest Laundering Paths

Traces flow routes between designated endpoints.



Ego Network Bounds Edge Reciprocity Scoring Interactive Node Expansion

COMPLIANCE & CONTROLS

Enterprise Governance & Security

Audit Trail

Automatic redaction of tokens and keys.

Production Guard

Hard blocks against accidental database reset.

Turnkey Deploy

Docker Compose ready with full test coverage.

Regulatory Guardrail: Decision-support triage scores explicitly separated from legal proof of guilt.

PROVEN IMPACT & ROADMAP

Triage Alerts in Under 5 Minutes

Slashing false positives on legitimate SMBs while capturing multi-hop rings in seconds.

90%
Alert Noise Reduction

12x
Faster SAR Preparation

100%
Explainable Attributions

Next: Real-Time Kafka Streaming • Temporal Graph Neural Networks (TGN) • FinCEN XML Export

Image Sources



https://images.stockcake.com/public/6/5/8/6587f152-dde0-4e3a-b244-521cde2a52f4_large/digital-financial-network-stockcake.jpg

Source: stockcake.com